Certificate of Conformity

9611 TELGE ROAD
HOUSTON, TX
77095
United States

Certificate Number:	HOU-2272501-1
Date:	April 23, 2025
Customer Name:	Cactus Wellhead LLC
Customer PO:	TRP0049827
Product:	718 11 SAAH
Part Number:	144712T REV A01
Cast Number:	LT7393EY11
Specification:	CMS-305 REV D

Comments:

Length: 223.875 in, 8pcs

This certificate confirms that the product(s) have been manufactured, sampled, tested, inspected and released in accordance with the requirements of the specification and purchase order, and meet the requirements.

Certified by:

Signature: *Digitally Signed By: R. Euton*

Name: Regina, Euton

Job Title: Certification Administrator

SPECIAL METALS
A PCC Company

HUNTINGTON ALLOYS CORPORATION
3200 Riverside Drive, Huntington, West Virginia 25705-1771 USA
Tel: +1.304.526.5100 Toll-Free in the USA: 1.800.334.4626
Fax: +1.304.526.5643 info@specialmetals.com

Certificate No. 146550-00
Dated 13-MAR-25
Page No. 1 / 9

CERTIFIED MATERIALS TEST REPORT

Note: The recording of false, fictitious or fraudulent statements or entries on this document may be punishable as a felony under federal statute.

This report relates only to the item(s) tested and may not be reproduced except in full.

Customer
HOWCO METALS MANAGEMENT LLC
9611 TELGE RD
HOUSTON
TX
77095

Ship To
HOWCO METALS MANAGEMENT LLC
9611 TELGE RD
HOUSTON
TX
77095

Sales Order Number	Purchase Order Number	Mark Order Number	Material Heat / Lot Identity
300042967 / 1.1	HOU-1038637-1	HOU-1038637-1	LT7393EY11

UNS Number
N07718

Material Description
INCONEL alloy 718, VACUUM INDUCTION MELTED-VACUUM ARC REMELTED, , HOT FIN ROUND, ROUGH T, SOL ANNEALED AND AGE HD, 11.0000, 72.0000, IN
1 PC PC# 1 6340 LBS

Specifications
HOWCO HG 718 BAR (GRADE 120K) REV 15 PER SMC COMMENTS / SMC HA 79 REV 10 (GRADE 120) / NACE MR0175(2021)-ISO 15156-3 (2020) / API 6ACRA (N07718) (120K) FIRST EDITION ERRATA 1, 2, & 3 ADDENDUM 1, 2, & 3 / API 6A718 2ND EDITION ERRATA 1 / SMC QCP 285 REV 3 / SMC MPP-718-LR REV 5.

ANALYSIS

	C %	MN %	FE %	S %	SI %	CU %	NI %	CR %	AL %
H	.015	.07	18.2	.0003	.06	.10	53.4	18.6	.52
Method	C/S	XRF	XRF	C/S	XRF	XRF	XRF	XRF	XRF

	TI %	MG %	CO %	MO %	NB %	TA %	B %	BI %	CA %
H	1.02	.0021	.14	2.88	5.02	.0071	.0034	<.000001	.0008
Method	XRF	AAS	XRF	XRF	XRF	XRF	OES	ICP-MS	OES

	P %	PB %	SE %	SN %
H	.004	<0.000001	<0.000001	.00047
Method	XRF	ICP-MS	ICP-MS	ICP-MS

	NB+TA
H	5.03

HOWCO
Mill Certificate Verified
11 APR 2025
Lesly de Leon Reyes
Quality Assurance Dept.
LRQA REG. No. 862242

INCOLOY®, INCONEL®, MONEL®, NILO®, NIMONIC®, Ni-SPAN-C®, UDIMET®, DURANICKEL®, 601GC®, 625LCF®, 718SPF®, 740H®, 800HT®, 945®, and 945X® are trademarks of the Special Metals group of companies

EF595

Customer PO#	Sales Order#	Howco Disp Cert#	Date	Authorized by	Page No.	Total Pages
TRP0049827	2272501-1	2272501-1	23-Apr-2025	Regina Euton	1	19

HUNTINGTON ALLOYS CORPORATION
3200 Riverside Drive, Huntington, West Virginia 25705-1771 USA
Tel: +1.304.626.5100 Toll-Free in the USA: 1.800.334.4626
Fax: +1.304.626.5643 info@specialmetals.com

Certificate No. 146550-00
Dated 13-MAR-25
Page No. 2 / 9

CERTIFIED MATERIALS TEST REPORT

Method

ANALYSIS METHOD LEGEND

AAS - Atomic absorption
C/S - Carbon/Sulfur
ICP - Inductively Coupled Plasma
ICP-MS - Inductively Coupled Plasma Mass Spectrometry
OES - Optical Emission Spectroscopy
XRF - X-Ray Fluorescence

TENSILE TEST

ROOM TEMP TENSILE - LONG|LAB|MECHANICAL

PIECE	TEST TEMPER	LOCATION	TENSILE KSI	.2% YIELD KSI	EFF GA LNGTH IN	RED OF AREA%	ELONG%	ORIENT
1	SG	H	181.2	135.8	2.000	44.8	27	LONG

HARDNESS TEST

HARDNESS - 120 SURVEY|LAB|MECHANICAL

Piece 1
Test Temper SG
Location H
Hard Scale HRC

120 Survey Individual

Position	Surface	Mid Radius	Center
2 Indent 1	38.4	38.4	39.2
2 Indent 2	37.8	38.7	40.0
2 Indent 3	37.5	38.3	40.1
6 Indent 1	37.6	38.5	
6 Indent 2	38.0	39.2	
6 Indent 3	37.9	38.3	
10 Indent 1	36.8	38.3	
10 Indent 2	37.0	39.0	
10 Indent 3	38.0	39.3	

120 Survey Average

Position	Surface	Mid Radius	Center
2 Oclock AVG	37.9	38.5	39.8
6 Oclock AVG	37.8	38.7	

INCOLOY®, INCONEL®, MONEL®, NILO®, NIMONIC®, Ni-SPAN-C®, UDIMET®, DURANICKEL®, 601GC®, 625LCF®, 718SPF®, 740H®, 800HT®, 945®, and 945X® are trademarks of the Special Metals group of companies

EF595

HUNTINGTON ALLOYS CORPORATION
3200 Riverside Drive, Huntington, West Virginia 25705-1771 USA
Tel: +1.304.526.5100 Toll-Free in the USA: 1.800.334.4626
Fax: +1.304.526.5643 info@specialmetals.com

Certificate No. 146550-00
Dated 13-MAR-25
Page No. 3 / 9

CERTIFIED MATERIALS TEST REPORT

10 Oclock AVG			37.3		38.9	

HARDNESS - STANDARD|LAB|MECHANICAL

TEST TEMPER	POSITION	HRD SCALE	VALUE 1	VALUE 2	VALUE 3	HRD AVG
SG	SUR	HRC	34.9	34.8	36.0	35.2

HEAT TREATMENT

TREATMENT	FURNACE	TEMP 1 F	TEMP 1 HOLD	HOLD 1 UNITS	CHANGE 1	CHANGE RATE 1
SOL ANN	SF2	1885	2	HR	WQ	IN:62/OUT:68

TREATMENT	FURNACE	TEMP 1 F	TEMP 1 HOLD	CHANGE 1
AGE HRD	SF2	1455	6HR/30MIN	AC

REMARK 1

HEAT TREAT CHARTS ATTACHED

OTHER TESTS

ETCH TEST (ORDER REQUIRED)|LAB|ETCH LAB

PIECE ID	TEST TEMPER	LOCATION	TEST VERDICT	REMARKS	ETCHANT DESCRIPTION
1	SG	H	P	TOTAL 2 ETCH/A604. 1 H AND T	Aqua Regia

ETCH TEST TO ASTM A604|LAB|ETCH LAB

PIECE ID	TEST TEMPER	LOCATION	ETCHANT DESCRIPTION	CLI RATING	CLII RATING	CLIII RATING	CLIV RATING
1	SG	H	Aqua Regia	A	A	A	A

GRAIN SIZE MEASUREMENT|LAB|METALLOGRAPHY

PIECE ID	TEST TEMPER	POSITION	TEST ORIENT	AV GS ASTM	NRM or DUP	GS CONFIG
1	SG	SUR	LONGITUDINAL	4	DUPLEX	WIDE RANGE

GS ASTM RNG 1	GS ASTM RNG 2	LOCATION
2	7	H

INCOLOY®, INCONEL®, MONEL®, NILO®, NIMONIC®, Ni-SPAN-C®, UDIMET®, DURANICKEL®, 601GC®, 625LCF®, 718SPF®, 740H®, 800HT®, 945®, and 945X® are trademarks of the Special Metals group of companies EF595

HUNTINGTON ALLOYS CORPORATION
3200 Riverside Drive, Huntington, West Virginia 25705-1771 USA
Tel: +1.304.526.5100 Toll-Free in the USA: 1.800.334.4626
Fax: +1.304.526.5643 info@specialmetals.com

Certificate No. 146550-00
Dated 13-MAR-25
Page No. 4 / 9

CERTIFIED MATERIALS TEST REPORT

PIECE ID	TEST TEMPER	POSITION	TEST ORIENT	AV GS ASTM	NRM or DUP	GS CONFIG
1	SG	MR	LONGITUDINAL	4	DUPLEX	WIDE RANGE

GS ASTM RNG 1	GS ASTM RNG 2	LOCATION
2	8	H

PIECE ID	TEST TEMPER	POSITION	TEST ORIENT	AV GS ASTM	NRM or DUP	GS CONFIG
1	SG	CEN	LONGITUDINAL	3	DUPLEX	WIDE RANGE

GS ASTM RNG 1	GS ASTM RNG 2	LOCATION
0	7	H

IMPACT CHARPY - TRANS|LAB|MECHANICAL

PIECE	TEST TEMPER	LOCATION	SIZE	STRIKER RAD SIZE	NOTCH ORIENT	J/SQ CM	FT LB	FTLB AVG
1	SG	H	10x10mm	8 mm	C-L	82.9	48.9	52.1

JOULES	TEMP°F	LAT EXP IN	%SHEAR
66.3	-75	.029	20

PIECE	TEST TEMPER	LOCATION	SIZE	STRIKER RAD SIZE	NOTCH ORIENT	J/SQ CM	FT LB	FTLB AVG
1	SG	H	10x10mm	8 mm	C-L	91.6	54.1	52.1

JOULES	TEMP°F	LAT EXP IN	%SHEAR
73.3	-75	.028	20

PIECE	TEST TEMPER	LOCATION	SIZE	STRIKER RAD SIZE	NOTCH ORIENT	J/SQ CM	FT LB	FTLB AVG
1	SG	H	10x10mm	8 mm	C-L	90.1	53.2	52.1

INCOLOY®, INCONEL®, MONEL®, NILO®, NIMONIC®, Ni-SPAN-C®, UDIMET®, DURANICKEL®, 601GC®, 625LCF®, 718SPF®, 740H®, 800HT®, 945®, and 945X® are trademarks of the Special Metals group of companies

EF595

HUNTINGTON ALLOYS CORPORATION
3200 Riverside Drive, Huntington, West Virginia 25705-1771 USA
Tel: +1.304.526.5100 Toll-Free in the USA: 1.800.334.4626
Fax: +1.304.526.5643 info@specialmetals.com

Certificate No. 146550-00
Dated 13-MAR-25
Page No. 5 / 9

CERTIFIED MATERIALS TEST REPORT

JOULES	TEMP°F	LAT EXP IN	%SHEAR
72.1	-75	.028	20

MICROSTRUCTURE TEST|LAB|METALLOGRAPHY

PIECE	TEST	TEMPER	LOCATION	TEST VERDICT	AREA EVAL	POSITION	METHOD	ETCHANT DESCRIPTION
1	SG		H	P	SURFACE	SUR	MMP 6A 718	Seven Acids

MAGNIFICATION
500X

PIECE	TEST	TEMPER	LOCATION	TEST VERDICT	AREA EVAL	POSITION	METHOD	ETCHANT DESCRIPTION
1	SG		H	P	MID RADIUS	MR	MMP 6A 718	Seven Acids

RATING	MAGNIFICATION
FIG A4	500X

PIECE	TEST	TEMPER	LOCATION	TEST VERDICT	AREA EVAL	POSITION	METHOD	ETCHANT DESCRIPTION
1	SG		H	P	CENTER	CEN	MMP 6A 718	Seven Acids

RATING	MAGNIFICATION
FIG A4	500X

LIST OF ATTACHMENTS

PHOTOS, HEAT TREAT CHARTS AND ULTRASONIC DATA SHEET ATTACHED

LOCATION LEGEND: B = BACK C = CENTER F = FRONT H = HEAD M = MIDDLE T = TOE

TEST VERDICT LEGEND: P = PASS W = WAIVER

TEST TEMPER LEGEND

SG - Solution anneal + aged

INCOLOY®, INCONEL®, MONEL®, NILO®, NIMONIC®, Ni-SPAN-C®, UDIMET®, DURANICKEL®, 601GC®, 625LCF®, 718SPF®, 740H®, 800HT®, 945®, and 945X® are trademarks of the Special Metals group of companies EF595

HUNTINGTON ALLOYS CORPORATION
3200 Riverside Drive, Huntington, West Virginia 25705-1771 USA
Tel: +1.304.526.5100 Toll-Free in the USA: 1.800.334.4626
Fax: +1.304.526.5643 info@specialmetals.com

Certificate No. 146550-00
Dated 13-MAR-25
Page No. 6 / 9

CERTIFIED MATERIALS TEST REPORT

CHEMISTRY IS REPORTED FROM REMELTED INGOT.
HEAT TREATMENT FURNACES ARE CALIBRATED TO AMS 2750, AS REQUIRED BY MIL-H-6875.

NO WELD REPAIR HAS BEEN PERFORMED ON THIS MATERIAL.
ACTUAL HOT WORKING REDUCTION RATIO: 4.7:1
REMELT INGOT SIZE: 24 IN

QUALIFICATION TEST COUPON WAS A PROLONGATION CUT FROM PRODUCTION MATERIAL AFTER HEAT TREATMENT.

FOR PRODUCT UP TO 12" DIAMETER, HEAT TREATMENT TIMES AND TEMPERATURES ARE NORMALLY MONITORED BY HEAT SINK THERMOCOUPLES, WITH LARGER DIAMETERS MONITORED BY CONTACT THERMOCOUPLES. ATTACHED FURNACE REPORTS DETAIL THE ACTUAL PRACTICE USED. WHEN HEAT SINK THERMOCOUPLES ARE USED, THE HEAT SINKS MEET API 6A REQUIREMENTS, BEING EQUAL TO OR UP TO 1/2" GREATER THAN THE LARGEST EQUIVALENT ROUND BAR WITHIN THE LOAD. FOR BAR DIAMETERS > 2" THE EMBEDDED THERMOCOUPLES SHALL BE >= 1" BELOW THE SURFACE OF THE HEAT SINKS. WHERE FURNACE LOADS CONTAIN BARS OF DIFFERENT DIAMETERS, THE MAXIMUM DIFFERENCE BETWEEN THE SMALLEST AND LARGEST BAR DIAMETER IS 2.0".

MATERIAL CAPABLE OF MEETING THE LIQUID PENETRANT REQUIREMENTS OF THE ORDER AFTER ADEQUATE SURFACE PREPARATION BY THE CUSTOMER TO PERFORM A VALID TEST.

ULTRASONICALLY TESTED AND ACCEPTABLE TO API 6A CLAUSES 10.4.2.12.2 AND 10.4.2.12.3 AND SATISFACTORY PER SMC QCP 285.

MICROSTRUCTURE SPECIMENS HAVE BEEN REMOVED FROM BAR PROLONGATIONS OBTAINED FROM EVERY HEAT TREAT LOT AFTER FINAL HEAT TREATMENT. A HEAT TREAT LOT CONSISTS OF BARS FROM THE SAME REMELT INGOT, OF THE SAME SIZE (EXCEPT LENGTH) AND PROCESSED IN THE SAME FURNACE LOADS FOR BOTH ANNEALING AND AGING.

MICROSTRUCTURE SPECIMENS AT NEAR SURFACE, MID-RADIUS, AND CENTER HAVE BEEN EXAMINED AT 100X AND 500X. NO TOPOLOGICAL DUPLEX GRAIN SIZE DISTRIBUTIONS PER ASTM E1181, I.E. BANDING, NECKLACE, OR CROSS-SECTION CONDITION, ARE PRESENT.
EXAMINATION FOR DELETERIOUS PHASES CONFIRMS FREEDOM FROM ACICULAR PHASES EXCEPT IN INDIVIDUAL, ISOLATED GRAINS NOT REPRESENTATIVE OF THE BULK MICROSTRUCTURE; NO INDIVIDUAL GRAINS ARE SURROUNDED BY ACICULAR PHASES.

INCOLOY®, INCONEL®, MONEL®, NILO®, NIMONIC®, Ni-SPAN-C®, UDIMET®, DURANICKEL®, 601GC®, 625LCF®, 718SPF®, 740H®, 800HT®, 945®, and 945X® are trademarks of the Special Metals group of companies

EF595

Customer PO#	Sales Order#	Howco Disp Cert#	Date	Authorized by	Page No.	Total Pages
TRP0049827	2272501-1	2272501-1	23-Apr-2025	Regina Euton	6	19

HUNTINGTON ALLOYS CORPORATION
3200 Riverside Drive, Huntington, West Virginia 25705-1771 USA
Tel: +1.304.526.5100 Toll-Free in the USA: 1.800.334.4626
Fax: +1.304.526.5643 info@specialmetals.com

Certificate No. 146550-00
Dated 13-MAR-25
Page No. 7 / 9

CERTIFIED MATERIALS TEST REPORT

MICROSTRUCTURE IS ALSO FREE FROM CONTINUOUS NETWORKS OF SECONDARY PHASES
ALONG GRAIN BOUNDARIES EXCEPT FOR INDIVIDUAL GRAINS OR ISOLATED FIELDS OF VIEW
NOT REPRESENTATIVE OF THE BULK MICROSTRUCTURE.
THE MICROSTRUCTURE IS FREE FROM LAVES PHASE.
DISCRETE, ISOLATED PARTICLES OF SECONDARY PHASES OR CARBIDES ARE ACCEPTABLE.

TOTAL OF ALL CHEMICAL ELEMENTS NOT REPORTED IS CONSIDERED RESIDUAL,
AND DOES NOT EXCEED. 0.50%

NIOBIUM (NB) EQUALS COLUMBIUM (CB).

THE MAXIMUM AND MINIMUM DIAMETERS IN THE HEAT TREAT BATCH ARE WITHIN
2 INCHES

MACROETCH SATISFACTORY AS PERFORMED AND EVALUATED PER API 6A718 / API 6ACRA; ASTM A604 SEVERITY A OR BETTER.

SMC-HA WAS THE TESTING FACILITY

SMC-HA WAS THE MELTING FACILITY

SMC-HA WAS THE HEAT TREATMENT FACILITY

SMC-HA WAS THE FINAL HOT WORKING FACILITY

FOR ROD AND BAR PRODUCTS, THE MATERIAL SIZE WHEN HEAT TREATED MAY BE
LARGER THAN THE SIZE SHIPPED DUE TO SUBSEQUENT PROCESSING.

FOR ALL PRODUCTS 3/4 INCH DIAMETER AND OVER, TENSILE TEST SPECIMENS
WERE 0.500 INCH GAUGE DIAMETER X 2 INCH GAUGE LENGTH AS STANDARD
SPECIMENS, WITH THE AXIS OF THE SPECIMEN LOCATION WITHIN THE PARENT
MATERIAL AT THE CENTER FOR PRODUCTS 1.5 INCH OR LESS OR AT
MID-RADIUS FOR PRODUCTS OVER 1.5 INCH IN ACCORDANCE WITH ASTM E8.
CHARPY IMPACT SPECIMENS FOR PRODUCT GREATER THAN 1.5" ARE TAKEN FROM MID RADIUS LOCATION.
FOR 1.5" OR LESS CHARPY IMPACT SPECIMENS ARE TAKEN FROM PRODUCT CENTER.

FOR BAR PRODUCTS LESS THAN ¾ INCH DIAMETER, TENSILE TEST SPECIMENS WERE .350" DIAMETER
SPECIMEN WITH A 1.4" GAGE LENGTH OR 4D WITH THE AXIS OF THE SPECIMEN LOCATION WITHIN
THE PARENT MATERIAL AT THE CENTER.

INCOLOY®, INCONEL®, MONEL®, NILO®, NIMONIC®, Ni-SPAN-C®, UDIMET®, DURANICKEL®, 601GC®, 625LCF®, 718SPF®, 740H®, 800HT®, 945®, and 945X® are trademarks of the Special Metals group of companies

EF595

HUNTINGTON ALLOYS CORPORATION
3200 Riverside Drive, Huntington, West Virginia 25705-1771 USA
Tel: +1.304.526.5100 Toll-Free in the USA: 1.800.334.4626
Fax: +1.304.526.5643 info@specialmetals.com

Certificate No. 146550-00
Dated 13-MAR-25
Page No. 8 / 9

CERTIFIED MATERIALS TEST REPORT

ALL TEST RESULTS ARE REPORTED TO AT LEAST THE REQUIRED PRECISION BY THE ROUNDING METHOD OF ASTM E 29
UNLESS OTHERWISE REQUIRED BY PURCHASE ORDER OR SPECIFICATION.

COUNTRY OF ORIGIN: MELTED AND MANUFACTURED IN THE USA. DFARS PART 252.225.7009 AND 252.225.7008 COMPLIANT
THIS CERTIFICATION AFFIRMS THAT THE CONTENTS OF THIS REPORT ARE CORRECT AND ACCURATE AND THAT ALL TEST RESULTS AND OPERATIONS PERFORMED BY SPECIAL METALS CORPORATION, INC. OR ITS SUBCONTRACTORS ARE IN COMPLIANCE WITH THE MATERIAL SPECIFICATIONS.
QUALITY SYSTEM MEETS REQUIREMENTS OF DIRECTIVE 2014/68/EU (PRESSURE EQUIPMENT DIRECTIVE),
ANNEX 1, CHAPTER 4.3 PER ABS GROUP LTD CERTIFICATE 41734 (EXPIRES JANUARY 28, 2027)

HUNTINGTON ALLOYS CORPORATION IS AN ACCREDITED INDEPENDENT NADCAP
MATERIALS TESTING LABORATORY VIA CERTIFICATE NUMBER 3200230925 EXPIRATION DATE: NOVEMBER 30, 2025
FOR ALL TESTING SPECIFIED IN THE SCOPE OF ACCREDITATION.
MATERIAL PRODUCED UNDER QA SYSTEM DOCUMENTED IN HUNTINGTON ALLOYS CORP QA MANUAL REV. 56, DATED 01/25/2024
EN 10 204 / DIN 50049 (TYPE 3.1)
QUALITY SYSTEM CERTIFICATION: AS9100D:2016, EN9100:2018, JISQ9100:2016 and ISO9001:2015
(ABS-QE CERT 56476 EXPIRES 1/28/2027)
LABORATORY IS ACCREDITED TO ISO/IEC 17025:2017 FOR MECHANICAL TESTING AND CHEMICAL ANALYSIS.

VISUAL AND DIMENSIONAL EXAMINATION SATISFACTORY.
MATERIAL, WHEN SHIPPED, IS FREE FROM CONTAMINATION BY MERCURY, RADIUM, ALPHA SOURCE, AND LOW MELTING ELEMENTS.

CHEMICAL ANALYSIS AS REQUIRED FOR CARBON, SULFUR, NITROGEN, OR OXYGEN IS PERFORMED BY COMBUSTION TECHNIQUES.
ALL OTHER REPORTED ELEMENTS ARE ANALYZED BY X-RAY AND/OR EMISSION SPECTROSCOPY.""

GE Transportation Aviation Special Process Certification Source Code 47150, Expiration Date of Dec 30, 2025

AUTHORIZED QUALITY CERTIFICATION REPRESENTATIVES:
D.L. CONLEY K.R. BREWER C.S. GUNNOE G.J. BURKHEAD T.A. WILKS

INCOLOY®, INCONEL®, MONEL®, NILO®, NIMONIC®, Ni-SPAN-C®, UDIMET®, DURANICKEL®, 601GC®, 625LCF®, 718SPF®, 740H®, 800HT®, 945®, and 945X® are trademarks of the Special Metals group of companies

EF595

Customer PO#	Sales Order#	Howco Disp Cert#	Date	Authorized by	Page No.	Total Pages
TRP0049827	2272501-1	2272501-1	23-Apr-2025	Regina Euton	8	19

SPECIAL METALS
A PCC Company

HUNTINGTON ALLOYS CORPORATION
3200 Riverside Drive, Huntington, West Virginia 25705-1771 USA
Tel: +1.304.526.5100 Toll-Free in the USA: 1.800.334.4626
Fax: +1.304.526.5643 info@specialmetals.com

Certificate No. 146550-00

Dated 13-MAR-25

CERTIFIED MATERIALS TEST REPORT

End Of Certificate

This is to certify that all required samplings inspections and tests have been performed in accordance with the order and specification requirements. The test report represents the actual attributes of the material furnished and the values shown are correct and true. The material described by this certificate is in full compliance with all order and inspection requirements. We hereby certify that the figures given are in accordance with the specified contract requirements.

REV. 8/08

Signed

For and on behalf of HUNTINGTON ALLOYS CORPORATION
Authorized Signature

INCOLOY®, INCONEL®, MONEL®, NILO®, NIMONIC®, Ni-SPAN-C®, UDIMET®, DURANICKEL®, 601GC®, 625LCF®, 718SPF®, 740H®, 800HT®, 945®, and 945X® are trademarks of the Special Metals group of companies

EF595

Customer PO#	Sales Order#	Howco Disp Cert#	Date	Authorized by	Page No.	Total Pages
TRP0049827	2272501-1	2272501-1	23-Apr-2025	Regina Euton	9	19

SPECIAL METALS

Ultrasonic Test Report

H-2245 Rev. 9/2023

Job Number	R3429671-02	**QC Procedure**	**QCP** 285 **Rev** 3 **Date** 03-03-2014
Heat Lot Number	LT7393EY11	**Additional Instruction**	**Supplement** **Rev** **Date** **Addendum** **Other** **Other**
Nom Diameter or X-section, wall or pipe schedule (in)	11.000"		
Number of Pieces **Piece Numbers if applied** **Serial Numbers if applied**	1 1	**Date of Inspection YYYY-MM-DD**	2025-03-07
Time of test, with respect to final length, width and heat-treat condition	Tested at final cross-section Tested prior to cutting final length / width Tested as final heat-treat condition	**Inspector and Certification Levels**	K. Gross SNT-TC-1ALevel II MIL-STD-2132 Inspector

Kevin Gross

Total Number of Probes Used for Test	1	**Shear Wave**				**Longitudinal Wave**	
		Circumferential Propagation		**Axial Propagation**		**Flaw Detection**	**Wall Thickness** ☐
		#1	**#2 Opposing**	**#1**	**#2 Opposing**		**Back Echo Monitor** ☐
						Contact Method	
Instrument Manufacture Model Serial Number Linearity Date m-d-y Frequency Filter Setting Cal Gain / Scan Gain dB		/	/	/	/	Staveley-Sonic 237 142PB 03-03-2025 2.25 MHz 50.0 / 56.0	/
Transducer Manufacture Model Serial Number Frequency MHz Element Size (in) Focal Length (in) Effective Beam (in) Refracted Angle Cable length/Type (ft) Reject On / Off		/	/	/	/	Single Element Krautkramer GAMMA RHP 13E026XD 2.25 1.0 dia 0 7.5 / RG/174/U OFF	/
Scanning Method Speed (ips) Index Type Max Index Used (in) Max Pulse Spacie (in) Couplant						Manual 6 Step Index Oil	
Reference Amplitudes % FSH % FSH Back Echo % Back Echo Loss EDAC Used						80 80 50 Yes	
Gating Delay and Range 1 Logic 1 % Ref Alarm Level 1 Delay and Range 2 Logic 2 % Ref Alarm Level 2						FS to BR Positive 50 BR Negative 50	
Reportable Conditions (Indications or BR loss)						None	
Remarks	1 WATER WASHABLE LP INSPECTION PERFORMED FOR UT SHEAR WAVE. 2. 3.						

Calibration Standard Data (all measurements in inches)

Calibration Standard		**NLOD Depth**	**NLID Depth**	**NTOD Depth**	**NTID Depth**	**FBH Diameter**	**Wall Thickness Reference 1**
SN	K367					0.250, 0.250, 0.250	
Standard Finish	No smoother than material	**NLOD Length**	**NLID Length**	**NTOD Length**	**NTID Length**	Metal Travel ☒ Drilled Depth ☐	**Wall Thickness Reference 2**
Material Finish	Less than 6.3 Micrometers					2.831, 5.567, 10.31	
Acoustic Comp Check	Long Wave Checked, 0 dB compensation required			Shear Wave , dB compensation required			

☐ **Minimum Wall Thickness Maintained** **Strip Chart Recorder Used** NA **Scanning Equipment**

☒ **Inspected Material Accepted** **Attachments**

"NOTE: The recording of false, fictitious or fraudulent statements or entries on this document may be punishable as a felony under Federal Statute."

SPECIAL METALS CORPORATION

3200 Riverside Drive

Huntington, WV «25707-1771 U.S.A.
304-526-5768 Cell 304-633-9143
www.specialmetals.com

TIMOTHY THACKER
NDE Manger

VISION RECORD

NONDESTRUCTIVE TESTING PERSONNEL

This is to certify that Kevin Gross has completed a vision screening which demonstrated the following results:

✓ **Natural** vision ability to read the Jaeger Number 1 line from the standard Jaeger test card at a distance of not less twelve inches.

___ **Corrected** vision ability to read Jaeger Number 1 line from the standard Jaeger test card at a distance of not less twelve inches.

Corrective aides used: a) glasses b) contacts c) both a and b d) either a or b

✓ Contrast discrimination **is sufficient** for requirements of NDT methods applied.

___ **Unable** to read Jaeger Number 1 line from the standard Jaeger test card at a distance of not less twelve inches. NDT certification at this time is not possible. Referred to eye care professional.

___ Contrast discrimination **is unacceptable** for requirements of NDT methods applied. NDT certification is not possible.

Date 01/27/2025

Timothy Thacker
NDE Manager,
Responsible Level III
ASNT NDT Level III UT, PT 216779
MIL-STD-2132 Test Examiner UT, PT, VT

NOTE: Ability to read the standard Jaeger "J1" test card at a distance of not less than twelve inches has been determined by the responsible Level III to be the equivalent of ability to achieve Snellen 20/25 rating.

INCONEL, INCOLOY, MONEL, NIMONIC, INCO-WELD, NILO, BRIGHTRAY, CORRONEL and UDIMET are registered trademarks of Special Metals Corporation

SPECIAL METALS
A PCC Company

Nondestructive Testing Personnel Certification

This is to certify that

Kevin Gross

has met all applicable training and examination requirements as outlined in MIL-STD-2132 and NSTR 99 and is therefore awarded certification as

Inspector, Ultrasonic – Immersion

Date of Certification: January 8, 2024

Date of Expiration: December 18, 2026
(Based on examination start date)

Vision Status: J1, Natural

T.P. Thacker,
Manager NDE
MIL-STD-2132 Test Examiner UT, PT, VT
ASNT NDT LEVEL III UT, PT 216779
SNT-TC-1A LEVEL III UT, PT

Corrective Aide:

Full documentation relating to this certification, which includes; training and experience histories, examination results and previous certification histories, is available for on-site review.

SPECIAL METALS
A PCC Company

Nondestructive Test Personnel Certification

This is to certify that

Kevin Gross

has met all applicable training and examination requirements as outlined in SMC written practice TQC 1 which meets or exceeds the requirements of SNT-TC-1A current and all prior editions, NAS 410 and T9074-AS-GIB-010/271.

Based upon training history, written, operational and physical examinations as required by the documents referenced above, this certification is awarded for

Level II, Ultrasonic-Immersion , Contact

Date of Issue: **January 8, 2024**

Date of Expiration: **December 18, 2026**

Vision Status: **J1, Natural**

T. Thacker Manager NDE
MIL-STD-2132 Test Examiner UT, PT, VT
ASNT NDT LEVEL III UT, PT 216776
SNT-TC-1A LEVEL III UT, PT

Correction Aid:

Documentation relating to this certification including examinations and previous certifications is available for on-site review.

SPECIAL METALS
A PCC Company

Nondestructive Test Personnel Certification

This is to certify that

Kevin Gross

has met all applicable training and examination requirements as outlined in MIL-STD-2132 and therefore, based on written, operational and physical examinations as specified in those documents referenced above, is awarded certification as

INSPECTOR, Visual, Dimensional and Alloy ID

Date of Issue: May 19, 2022

Date of Expiration: May 19, 2025

Vision Status: J1, Natural

T. Thacker Manager NDE
MIL-STD-2132 Test Examiner UT, PT, VT
ASNT NDT LEVEL III UT, PT 216776
SNT-TC-1A LEVEL III UT, PT

Documentation relating to this certification including examinations and previous certifications is available for on-site review.

Correction Aid: None

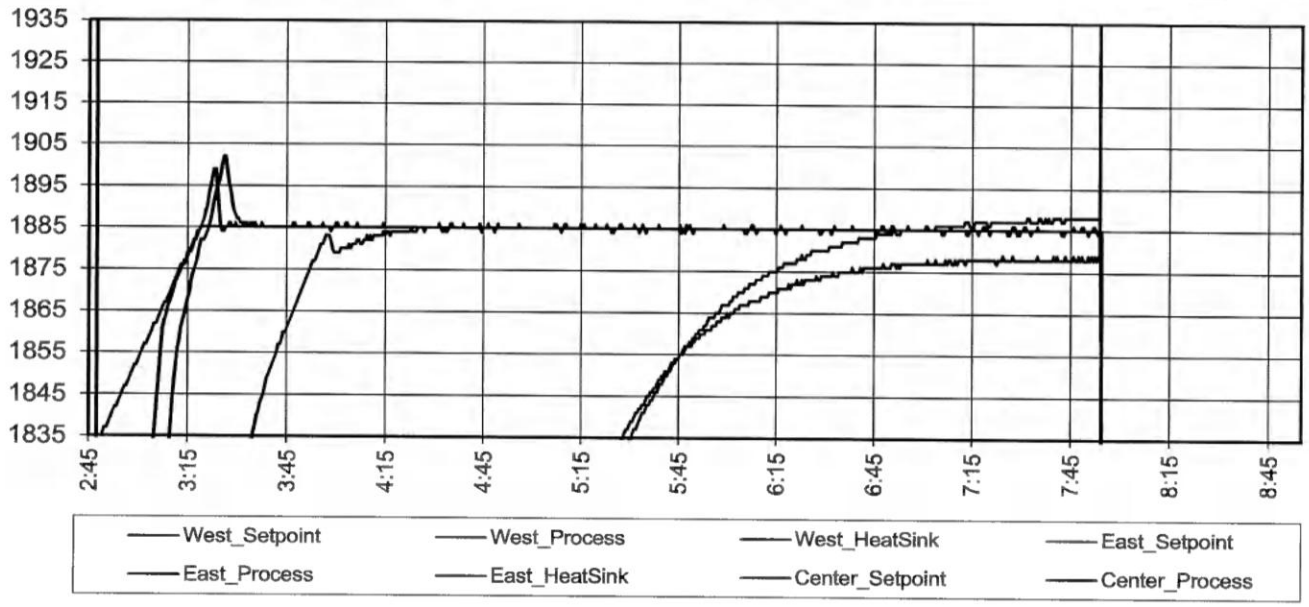
Customer PO#	Sales Order#	Howco Disp Cert#	Date	Authorized by	Page No.	Total Pages
TRP0049827	2272501-1	2272501-1	23-Apr-2025	Regina Euton	14	19

Order Report - Machine Shop Tip-Up Furnace #2 (50 Ft.)

Date & Time:	3/6/25	2:47:07
Operator:	13608	
Heat Type:	Solution Anneal	
Recipe #	167	
Quench:	Yes	
Water Temp (Start/End):		62 / 68
Maximum Water Temp:		68

Heat Sink Information	
East Heat Lot#	059134-24
East Heat Offset	1 °F
East heat Sink Size	Contact T/C
West Heat Lot#	083474-24
West Heat Offset	3 °F
West Heat Sink Size	Contact T/C

	Recipe Information			Event Information			
Event	Temp. SP	Hold Time (In Min.)	Ramp Rate (Per Hr.)	Start Date	Start Time	End Date	End Time
Charge				3/6/2025	2:47:07		
Step #1	1,885 °F	120	0 °F	3/6/2025	5:52:42	3/6/2025	7:52:42
Step #2	0 °F	0	0 °F				
Step #3	0 °F	0	0 °F				
Step #4	0 °F	0					
Discharge				3/6/2025	7:54:27		
Quench Time	60	Start Recipe HS Offset	25 °F	3/6/2025	7:54:49	3/6/2025	8:54:53



Order	Heat	Lot	Alloy	Weight	Dim1	Dim2	Dim3	Pieces	Location
R3429671-02	LT7393EY	11-1	718	7500	0	11	266.062	1	C11

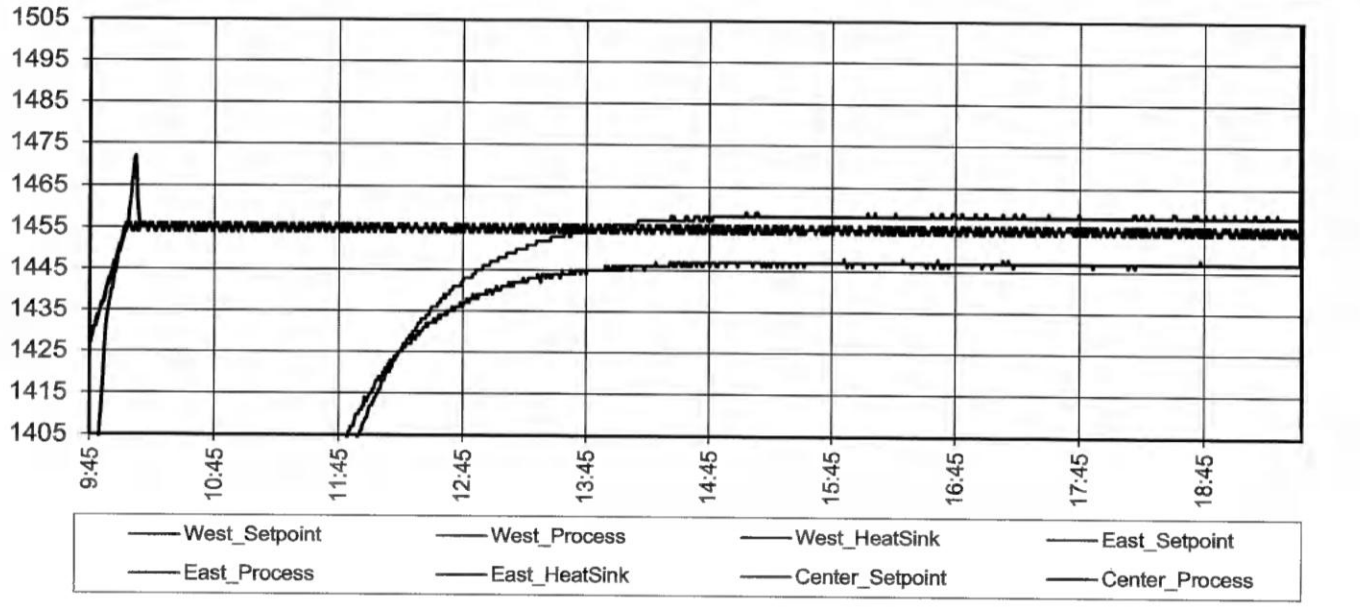
Customer PO#	Sales Order#	Howco Disp Cert#	Date	Authorized by	Page No.	Total Pages
TRP0049827	2272501-1	2272501-1	23-Apr-2025	Regina Euton	15	19

Order Report - Machine Shop Tip-Up Furnace #2 (50 Ft.)

Date & Time:	3/6/25	9:30:34
Operator:	13921	
Heat Type:	Age Hard	
Recipe #	132	
Quench:	No	
Water Temp (Start/End):		

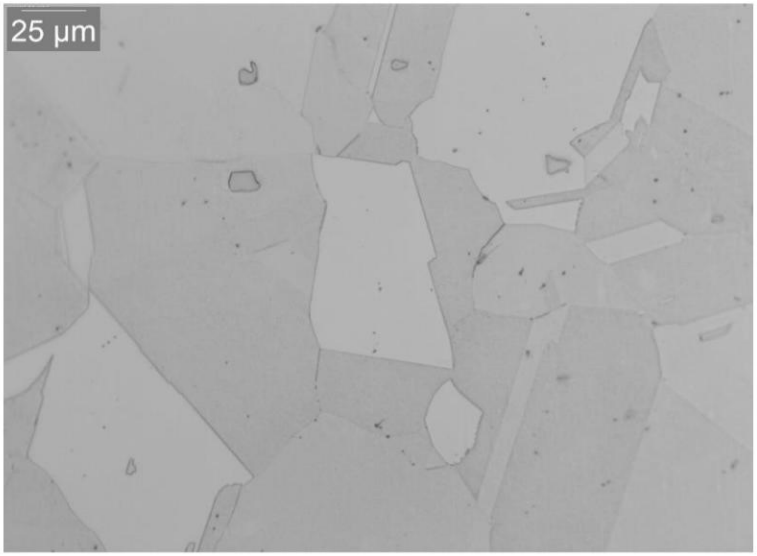
Heat Sink Information	
East Heat Lot#	059134-24
East Heat Offset	°F
East heat Sink Size	Contact T/C
West Heat Lot#	083474-24
West Heat Offset	2 °F
West Heat Sink Size	Contact T/C

	Recipe Information			Event Information			
Event	Temp. SP	Hold Time (In Min.)	Ramp Rate (Per Hr.)	Start Date	Start Time	End Date	End Time
Charge				3/6/2025	9:30:34		
Step #1	1,455 °F	390	0 °F	3/6/2025	12:59:54	3/6/2025	19:29:53
Step #2	0 °F	0	0 °F				
Step #3	0 °F	0	0 °F				
Step #4	0 °F	0					
Discharge				3/6/2025	19:31:38		
Quench Time		Start Recipe HS Offset	15 °F				



Order	Heat	Lot	Alloy	Weight	Dim1	Dim2	Dim3	Pieces	Location
R3429671-02	LT7393EY	11-1	718	7500	0	11	266.062	1	C11

Customer PO#	Sales Order#	Howco Disp Cert#	Date	Authorized by	Page No.	Total Pages
TRP0049827	2272501-1	2272501-1	23-Apr-2025	Regina Euton	16	19



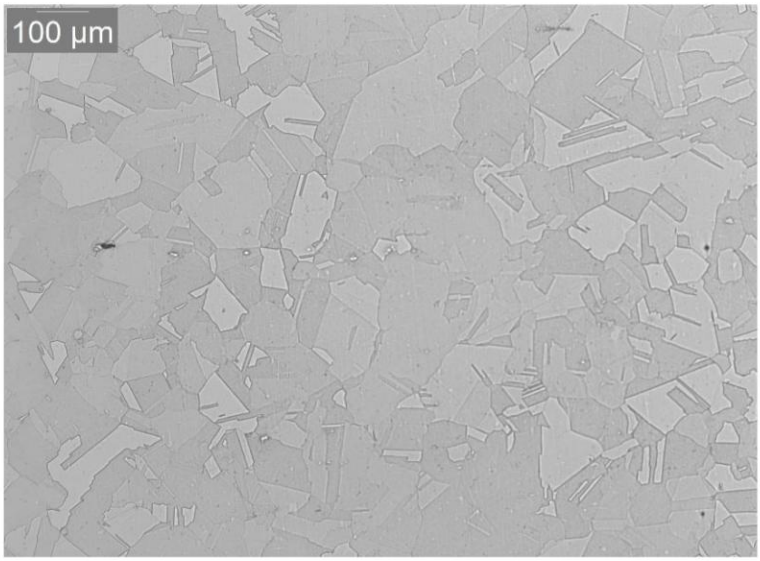


Figure 1: MS24754

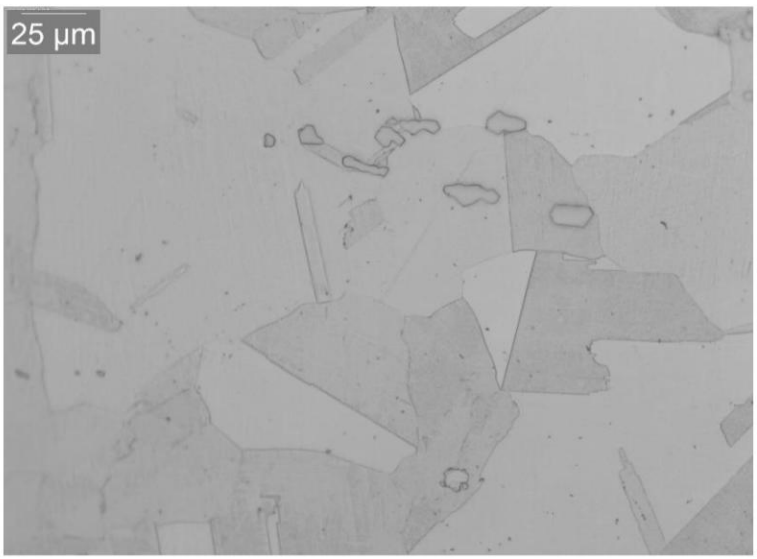
Figure 2: GS24755

SPECIAL METALS

Sales Order Number	300042967-1
Heat Number/Lot No.	LT7393EY11
Piece I.D.	1 HEAD
Alloy	INCONEL alloy 718
Product Form	Round
Product Size	11.0000”
Product Condition	Solution Ann and Aged
Lab Heat Treat	None
Image Number	See Photomicrograph
Magnification	500x-100x
Orientation	Long-Cen
Etchant	7-Acids
Etch Time	Standard

NOTE: The recording of false, fictitious or fraudulent statements or entries on this document may be punishable as a felony under Federal Statute.

Customer PO#	Sales Order#	Howco Disp Cert#	Date	Authorized by	Page No.	Total Pages
TRP0049827	2272501-1	2272501-1	23-Apr-2025	Regina Euton	17	19



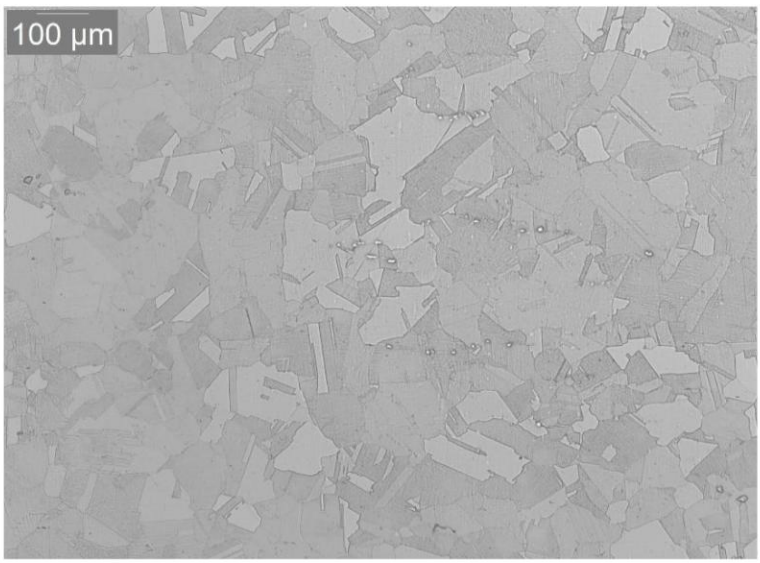


Figure 1: MS24752

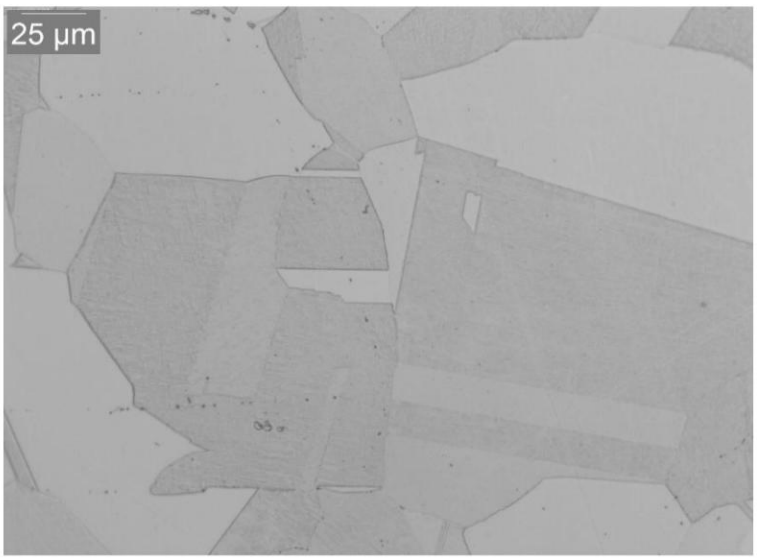
Figure 2: GS24753

SPECIAL METALS

Sales Order Number	300042967-1
Heat Number/Lot No.	LT7393EY11
Piece I.D.	1 HEAD
Alloy	INCONEL alloy 718
Product Form	Round
Product Size	11.0000”
Product Condition	Solution Ann and Aged
Lab Heat Treat	None
Image Number	See Photomicrograph
Magnification	500x-100x
Orientation	Long-Mr
Etchant	7-Acids
Etch Time	Standard

NOTE: The recording of false, fictitious or fraudulent statements or entries on this document may be punishable as a felony under Federal Statute.

Customer PO#	Sales Order#	Howco Disp Cert#	Date	Authorized by	Page No.	Total Pages
TRP0049827	2272501-1	2272501-1	23-Apr-2025	Regina Euton	18	19



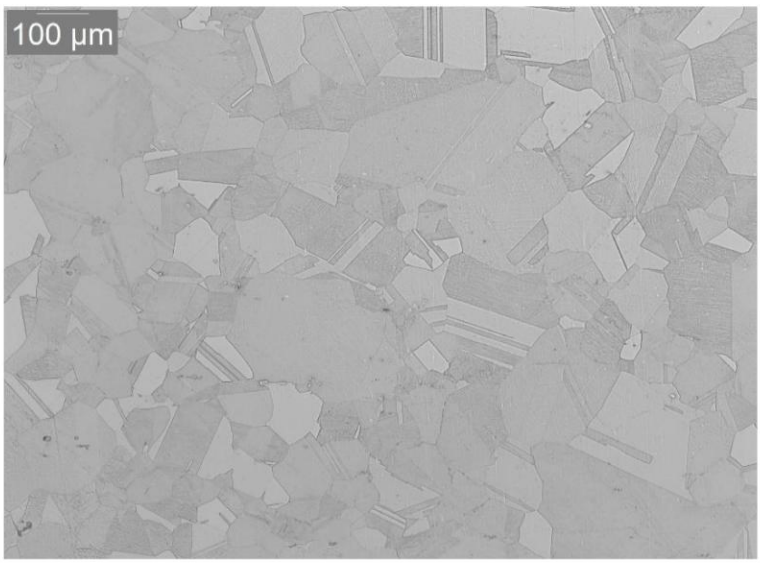


Figure 1: MS24750

Figure 2: GS24751

SPECIAL METALS

Sales Order Number	300042967-1
Heat Number/Lot No.	LT7393EY11
Piece I.D.	1 HEAD
Alloy	INCONEL alloy 718
Product Form	Round
Product Size	11.0000”
Product Condition	Solution Ann and Aged
Lab Heat Treat	None
Image Number	See Photomicrograph
Magnification	500x-100x
Orientation	Long-Sur
Etchant	7-Acids
Etch Time	Standard

NOTE: The recording of false, fictitious or fraudulent statements or entries on this document may be punishable as a felony under Federal Statute.

Customer PO#	Sales Order#	Howco Disp Cert#	Date	Authorized by	Page No.	Total Pages
TRP0049827	2272501-1	2272501-1	23-Apr-2025	Regina Euton	19	19

Certificate of Conformity

9611 TELGE ROAD
HOUSTON, TX
77095
United States

Certificate Number:	HOU-2272501-1
Date:	April 23, 2025
Customer Name:	Cactus Wellhead LLC
Customer PO:	TRP0049827
Product:	718 11 SAAH
Part Number:	144712T REV A01
Cast Number:	LT7386EY11
Specification:	CMS-305 REV D

Comments:

Length: 233.00 in, 5pcs

This certificate confirms that the product(s) have been manufactured, sampled, tested, inspected and released in accordance with the requirements of the specification and purchase order, and meet the requirements.

Certified by:

Signature: *Digitally Signed By: R. Euton*

Name: Regina, Euton

Job Title: Certification Administrator

HUNTINGTON ALLOYS CORPORATION
3200 Riverside Drive, Huntington, West Virginia 25705-1771 USA
Tel: +1.304.526.5100 Toll-Free in the USA: 1.800.334.4626
Fax: +1.304.526.5643 info@specialmetals.com

Certificate No. 146118-00
Dated 25-FEB-25
Page No. 1 / 8

CERTIFIED MATERIALS TEST REPORT

Note: The recording of false, fictitious or fraudulent statements or entries on this document may be punishable as a felony under federal statute.

This report relates only to the item(s) tested and may not be reproduced except in full.

Customer
HOWCO METALS MANAGEMENT LLC
9611 TELGE RD
HOUSTON
TX
77095

Ship To
HOWCO METALS MANAGEMENT LLC
9611 TELGE RD
HOUSTON
TX
77095

Sales Order Number	Purchase Order Number	Mark Order Number	Material Heat / Lot Identity
300042919 / 1.1	HOU-1038175-1	HOU-1038175-1	LT7386EY11

UNS Number
N07718

Material Description

INCONEL alloy 718, VACUUM INDUCTION MELTED-VACUUM ARC REMELTED, , HOT FIN ROUND, ROUGH T, SOL ANNEALED AND AGE HD, 11.0000, 72.0000, IN
1 PC PC# 1 6620 LBS

Specifications

HOWCO HG 718 BAR (GRADE 120K) REV 15 PER SMC COMMENTS / SMC HA 79 REV 10 (GRADE 120) / NACE MR0175(2021)-ISO 15156-3 (2020) / API 6ACRA (N07718) (120K) FIRST EDITION ERRATA 1, 2, & 3 ADDENDUM 1, 2, & 3 / API 6A718 2ND EDITION ERRATA 1 / SMC QCP 285 REV 3 / SMC MPP-718-LR REV 4.

ANALYSIS

	C %	MN %	FE %	S %	SI %	CU %	NI %	CR %	AL %
H	.015	.10	18.1	.001	.07	.07	53.4	18.4	.51
Method	C/S	XRF	XRF	C/S	XRF	XRF	XRF	XRF	XRF

	TI %	MG %	CO %	MO %	NB %	TA %	B %	BI %	CA %
H	1.01	.0026	.26	2.90	5.07	.0081	.0037	<.000001	.0005
Method	XRF	AAS	XRF	XRF	XRF	XRF	OES	ICP-MS	OES

	P %	PB %	SE %	SN %
H	.007	<0.000001	<0.000001	.00060
Method	XRF	ICP-MS	ICP-MS	ICP-MS

	NB+TA
H	5.07

3/10/2025, 3:56:23 pm
HOWCO
Mill Certificate Verified
Harley Brewer
Quality Assurance Dept.
IRQA REG. No 862242

INCOLOY®, INCONEL®, MONEL®, NILO®, NIMONIC®, Ni-SPAN-C®, UDIMET®, DURANICKEL®, 601GC®, 625LCF®, 718SPF®, 740H®, 800HT®, 945®, and 945X® are trademarks of the Special Metals group of companies

EF595

Customer PO#	Sales Order#	Howco Disp Cert#	Date	Authorized by	Page No.	Total Pages
TRP0049827	2272501-1	2272501-1	23-Apr-2025	Regina Euton	1	18

HUNTINGTON ALLOYS CORPORATION
3200 Riverside Drive, Huntington, West Virginia 25705-1771 USA
Tel: +1.304.526.5100 Toll-Free in the USA: 1.800.334.4626
Fax: +1.304.526.5643 info@specialmetals.com

Certificate No. 146118-00
Dated 25-FEB-25
Page No. 2 / 8

CERTIFIED MATERIALS TEST REPORT

Method

ANALYSIS METHOD LEGEND

AAS - Atomic absorption
C/S - Carbon/Sulfur
ICP - Inductively Coupled Plasma
ICP-MS - Inductively Coupled Plasma Mass Spectrometry
OES - Optical Emission Spectroscopy
XRF - X-Ray Fluorescence

TENSILE TEST

ROOM TEMP TENSILE - LONG|LAB|MECHANICAL

PIECE	TEST TEMPER	LOCATION	TENSILE KSI	.2% YIELD KSI	EFF GA LNGTH IN	RED OF AREA%	ELONG%	ORIENT
1	SG	T	181.2	137.9	2.000	41.7	26.5	LONG

HARDNESS TEST

HARDNESS - 120 SURVEY|LAB|MECHANICAL

Piece 1
Test Temper SG
Location T
Hard Scale HRC

120 Survey Individual

Position	Surface	Mid Radius	Center
2 Indent 1	36.9	39.4	38.1
2 Indent 2	37.3	39.9	38.9
2 Indent 3	38.1	39.7	39.0
6 Indent 1	38.5	38.1	
6 Indent 2	39.5	37.7	
6 Indent 3	38.8	37.8	
10 Indent 1	38.4	39.0	
10 Indent 2	38.3	39.9	
10 Indent 3	38.1	39.4	

120 Survey Average

Position	Surface	Mid Radius	Center
2 Oclock AVG	37.4	39.7	38.7
6 Oclock AVG	38.9	37.9	

INCOLOY®, INCONEL®, MONEL®, NILO®, NIMONIC®, Ni-SPAN-C®, UDIMET®, DURANICKEL®, 601GC®, 625LCF®, 718SPF®, 740H®, 800HT®, 945®, and 945X® are trademarks of the Special Metals group of companies EF595

HUNTINGTON ALLOYS CORPORATION
3200 Riverside Drive, Huntington, West Virginia 25705-1771 USA
Tel: +1.304.526.5100 Toll-Free in the USA: 1.800.334.4626
Fax: +1.304.526.5643 info@specialmetals.com

Certificate No. 146118-00
Dated 25-FEB-25
Page No. 3 / 8

CERTIFIED MATERIALS TEST REPORT

10 Oclock AVG	38.3	39.4		

HARDNESS - STANDARD|LAB|MECHANICAL

TEST TEMPER	POSITION	HRD SCALE	VALUE 1	VALUE 2	VALUE 3	HRD AVG
SG	SUR	HRC	33.9	35.3	34.8	34.7

HEAT TREATMENT

TREATMENT	FURNACE	TEMP 1 F	TEMP 1 HOLD	HOLD 1 UNITS	CHANGE 1	CHANGE RATE 1
SOL ANN	SF2	1885	2	HR	WQ	IN:60/OUT:67

TREATMENT	FURNACE	TEMP 1 F	TEMP 1 HOLD	CHANGE 1
AGE HRD	SF2	1455	6HR/30MIN	AC

REMARK 1

HEAT TREAT CHARTS ATTACHED

OTHER TESTS

ETCH TEST (ORDER REQUIRED)|LAB|ETCH LAB

PIECE ID	TEST TEMPER	LOCATION	TEST VERDICT	REMARKS	ETCHANT DESCRIPTION
1	SG	H	P	TOTAL 2 ETCH/A604 1H AND T	Aqua Regia

ETCH TEST TO ASTM A604|LAB|ETCH LAB

PIECE ID	TEST TEMPER	LOCATION	ETCHANT DESCRIPTION	CLI RATING	CLII RATING	CLIII RATING	CLIV RATING
1	SG	H	Aqua Regia	A	A	A	A

GRAIN SIZE MEASUREMENT|LAB|METALLOGRAPHY

PIECE ID	TEST TEMPER	POSITION	TEST ORIENT	AV GS ASTM	NRM or DUP	LOCATION
1	SG	SUR	LONGITUDINAL	2.5	NORMAL	T
1	SG	MR	LONGITUDINAL	4.5	NORMAL	T
1	SG	CEN	LONGITUDINAL	4	NORMAL	T

IMPACT CHARPY - TRANS|LAB|MECHANICAL

PIECE	TEST TEMPER	LOCATION	SIZE	STRIKER RAD SIZE	NOTCH ORIENT	J/SQ CM	FT LB	FTLB AVG

INCOLOY®, INCONEL®, MONEL®, NILO®, NIMONIC®, Ni-SPAN-C®, UDIMET®, DURANICKEL®, 601GC®, 625LCF®, 718SPF®, 740H®, 800HT®, 945®, and 945X® are trademarks of the Special Metals group of companies EF595

Customer PO#	Sales Order#	Howco Disp Cert#	Date	Authorized by	Page No.	Total Pages
TRP0049827	2272501-1	2272501-1	23-Apr-2025	Regina Euton	3	18

SPECIAL METALS
A PCC Company

HUNTINGTON ALLOYS CORPORATION
3200 Riverside Drive, Huntington, West Virginia 25705-1771 USA
Tel: +1.304.526.5100 Toll-Free in the USA: 1.800.334.4626
Fax: +1.304.526.5643 info@specialmetals.com

CERTIFIED MATERIALS TEST REPORT

1	SG	T	10x10mm	8 mm	C-L	80.4	47.4	48.3

JOULES	TEMP°F	LAT EXP IN	%SHEAR
64.3	-75	.027	20

PIECE	TEST	TEMPER	LOCATION	SIZE	STRIKER RAD SIZE	NOTCH ORIENT	J/SQ CM	FT LB	FTLB AVG
1	SG	T	10x10mm	8 mm		C-L	85.6	50.5	48.3

JOULES	TEMP°F	LAT EXP IN	%SHEAR
68.5	-75	.025	20

PIECE	TEST	TEMPER	LOCATION	SIZE	STRIKER RAD SIZE	NOTCH ORIENT	J/SQ CM	FT LB	FTLB AVG
1	SG	T	10x10mm	8 mm		C-L	79.5	46.9	48.3

JOULES	TEMP°F	LAT EXP IN	%SHEAR
63.6	-75	.026	20

MICROSTRUCTURE TEST|LAB|METALLOGRAPHY

PIECE	TEST	TEMPER	LOCATION	TEST VERDICT	AREA EVAL	POSITION	METHOD	ETCHANT DESCRIPTION
1	SG	T	P		SURFACE	SUR	MMP 6A 718	Seven Acids

MAGNIFICATION
500X

PIECE	TEST	TEMPER	LOCATION	TEST VERDICT	AREA EVAL	POSITION	METHOD	ETCHANT DESCRIPTION
1	SG	T	P		MID RADIUS	MR	MMP 6A 718	Seven Acids

RATING	MAGNIFICATION

INCOLOY®, INCONEL®, MONEL®, NILO®, NIMONIC®, Ni-SPAN-C®, UDIMET®, DURANICKEL®, 601GC®, 625LCF®, 718SPF®, 740H®, 800HT®, 945®, and 945X® are trademarks of the Special Metals group of companies EF595

HUNTINGTON ALLOYS CORPORATION
3200 Riverside Drive, Huntington, West Virginia 26705-1771 USA
Tel: +1.304.526.5100 Toll-Free in the USA: 1.800.334.4626
Fax: +1.304.526.5643 info@specialmetals.com

Certificate No. 146118-00
Dated 25-FEB-25

CERTIFIED MATERIALS TEST REPORT

FIG.A.3 500X

PIECE	TEST TEMPER	LOCATION	TEST VERDICT	AREA EVAL	POSITION	METHOD	ETCHANT DESCRIPTION
1	SG	T	P	CENTER	CEN	MMP 6A 718	Seven Acids

RATING	MAGNIFICATION
FIG.A.3	500X

LIST OF ATTACHMENTS

HEAT TREAT CHARTS, PHOTOS, AND ULTRASONIC DATA SHEET ATTACHED

LOCATION LEGEND: B = BACK C = CENTER F = FRONT H = HEAD M = MIDDLE T = TOE

TEST VERDICT LEGEND: P = PASS W = WAIVER
TEST TEMPER LEGEND
SG - Solution anneal + aged
CHEMISTRY IS REPORTED FROM REMELTED INGOT.
HEAT TREATMENT FURNACES ARE CALIBRATED TO AMS 2750, AS REQUIRED BY MIL-H-6875.

NO WELD REPAIR HAS BEEN PERFORMED ON THIS MATERIAL.
ACTUAL HOT WORKING REDUCTION RATIO: 4.8:1
REMELT INGOT SIZE: 24 IN

QUALIFICATION TEST COUPON WAS A PROLONGATION CUT FROM PRODUCTION MATERIAL AFTER HEAT TREATMENT.

FOR PRODUCT UP TO 12" DIAMETER, HEAT TREATMENT TIMES AND TEMPERATURES ARE NORMALLY MONITORED BY HEAT SINK THERMOCOUPLES, WITH LARGER DIAMETERS MONITORED BY CONTACT THERMOCOUPLES. ATTACHED FURNACE REPORTS DETAIL THE ACTUAL PRACTICE USED. WHEN HEAT SINK THERMOCOUPLES ARE USED, THE HEAT SINKS MEET API 6A REQUIREMENTS, BEING EQUAL TO OR UP TO 1/2" GREATER THAN THE LARGEST EQUIVALENT ROUND BAR WITHIN THE LOAD. FOR BAR DIAMETERS > 2" THE EMBEDDED THERMOCOUPLES SHALL BE >= 1" BELOW THE SURFACE OF THE HEAT SINKS. WHERE FURNACE LOADS CONTAIN BARS OF DIFFERENT DIAMETERS, THE MAXIMUM DIFFERENCE BETWEEN THE SMALLEST AND LARGEST BAR DIAMETER IS 2.0".

INCOLOY®, INCONEL®, MONEL®, NILO®, NIMONIC®, Ni-SPAN-C®, UDIMET®, DURANICKEL®, 601GC®, 625LCF®, 718SPF®, 740H®, 800HT®, 945®, and 945X® are trademarks of the Special Metals group of companies EF595

HUNTINGTON ALLOYS CORPORATION
3200 Riverside Drive, Huntington, West Virginia 25705-1771 USA
Tel: +1.304.526.5100 Toll-Free in the USA: 1.800.334.4626
Fax: +1.304.526.5643 info@specialmetals.com

Certificate No. 146118-00
Dated 25-FEB-25

CERTIFIED MATERIALS TEST REPORT

MATERIAL CAPABLE OF MEETING THE LIQUID PENETRANT REQUIREMENTS OF THE
ORDER AFTER ADEQUATE SURFACE PREPARATION BY THE CUSTOMER TO PERFORM
A VALID TEST.

ULTRASONICALLY TESTED AND ACCEPTABLE TO API 6A CLAUSES 10.4.2.12.2 AND
10.4.2.12.3 AND SATISFACTORY PER SMC QCP 285.

MICROSTRUCTURE SPECIMENS HAVE BEEN REMOVED FROM BAR PROLONGATIONS
OBTAINED FROM EVERY HEAT TREAT LOT AFTER FINAL HEAT TREATMENT. A HEAT
TREAT LOT CONSISTS OF BARS FROM THE SAME REMELT INGOT, OF THE SAME
SIZE (EXCEPT LENGTH) AND PROCESSED IN THE SAME FURNACE LOADS FOR BOTH
ANNEALING AND AGING.

MICROSTRUCTURE SPECIMENS AT NEAR SURFACE, MID-RADIUS, AND CENTER HAVE BEEN EXAMINED
AT 100X AND 500X. NO TOPOLOGICAL DUPLEX GRAIN SIZE DISTRIBUTIONS PER ASTM E1181,
I.E. BANDING, NECKLACE, OR CROSS-SECTION CONDITION, ARE PRESENT.
EXAMINATION FOR DELETERIOUS PHASES CONFIRMS FREEDOM FROM ACICULAR PHASES
EXCEPT IN INDIVIDUAL, ISOLATED GRAINS NOT REPRESENTATIVE OF THE BULK MICROSTRUCTURE;
NO INDIVIDUAL GRAINS ARE SURROUNDED BY ACICULAR PHASES.
MICROSTRUCTURE IS ALSO FREE FROM CONTINUOUS NETWORKS OF SECONDARY PHASES
ALONG GRAIN BOUNDARIES EXCEPT FOR INDIVIDUAL GRAINS OR ISOLATED FIELDS OF VIEW
NOT REPRESENTATIVE OF THE BULK MICROSTRUCTURE.
THE MICROSTRUCTURE IS FREE FROM LAVES PHASE.
DISCRETE, ISOLATED PARTICLES OF SECONDARY PHASES OR CARBIDES ARE ACCEPTABLE.

TOTAL OF ALL CHEMICAL ELEMENTS NOT REPORTED IS CONSIDERED RESIDUAL,
AND DOES NOT EXCEED. 0.50%

NIOBIUM (NB) EQUALS COLUMBIUM (CB).

THE MAXIMUM AND MINIMUM DIAMETERS IN THE HEAT TREAT BATCH ARE WITHIN
2 INCHES

MACROETCH SATISFACTORY AS PERFORMED AND EVALUATED PER API 6A718 / API 6ACRA; ASTM A604 SEVERITY A OR BETTER.

SMC-HA WAS THE TESTING FACILITY

SMC-HA WAS THE MELTING FACILITY

INCOLOY®, INCONEL®, MONEL®, NILO®, NIMONIC®, Ni-SPAN-C®, UDIMET®, DURANICKEL®, 601GC®, 625LCF®, 718SPF®, 740H®, 800HT®, 945®, and 945X® are trademarks of the Special Metals group of companies

Customer PO#	Sales Order#	Howco Disp Cert#	Date	Authorized by	Page No.	Total Pages
TRP0049827	2272501-1	2272501-1	23-Apr-2025	Regina Euton	6	18

HUNTINGTON ALLOYS CORPORATION
3200 Riverside Drive, Huntington, West Virginia 25705-1771 USA
Tel: +1.304.526.5100 Toll-Free in the USA: 1.800.334.4626
Fax: +1.304.526.5643 info@specialmetals.com

Certificate No. 146118-00
Dated 25-FEB-25

CERTIFIED MATERIALS TEST REPORT

SMC-HA WAS THE HEAT TREATMENT FACILITY

SMC-HA WAS THE FORGING FACILITY

FOR ROD AND BAR PRODUCTS, THE MATERIAL SIZE WHEN HEAT TREATED MAY BE LARGER THAN THE SIZE SHIPPED DUE TO SUBSEQUENT PROCESSING.

FOR ALL PRODUCTS 3/4 INCH DIAMETER AND OVER, TENSILE TEST SPECIMENS WERE 0.500 INCH GAUGE DIAMETER X 2 INCH GAUGE LENGTH AS STANDARD SPECIMENS, WITH THE AXIS OF THE SPECIMEN LOCATION WITHIN THE PARENT MATERIAL AT THE CENTER FOR PRODUCTS 1.5 INCH OR LESS OR AT MID-RADIUS FOR PRODUCTS OVER 1.5 INCH IN ACCORDANCE WITH ASTM E8.
CHARPY IMPACT SPECIMENS FOR PRODUCT GREATER THAN 1.5" ARE TAKEN FROM MID RADIUS LOCATION.
FOR 1.5" OR LESS CHARPY IMPACT SPECIMENS ARE TAKEN FROM PRODUCT CENTER.

FOR BAR PRODUCTS LESS THAN ¾ INCH DIAMETER, TENSILE TEST SPECIMENS WERE .350" DIAMETER SPECIMEN WITH A 1.4" GAGE LENGTH OR 4D WITH THE AXIS OF THE SPECIMEN LOCATION WITHIN THE PARENT MATERIAL AT THE CENTER.

ALL TEST RESULTS ARE REPORTED TO AT LEAST THE REQUIRED PRECISION BY THE ROUNDING METHOD OF ASTM E 29 UNLESS OTHERWISE REQUIRED BY PURCHASE ORDER OR SPECIFICATION.

COUNTRY OF ORIGIN: MELTED AND MANUFACTURED IN THE USA. DFARS PART 252.225.7009 AND 252.225.7008 COMPLIANT
THIS CERTIFICATION AFFIRMS THAT THE CONTENTS OF THIS REPORT ARE CORRECT AND ACCURATE AND THAT ALL TEST RESULTS AND OPERATIONS PERFORMED BY SPECIAL METALS CORPORATION, INC. OR ITS SUBCONTRACTORS ARE IN COMPLIANCE WITH THE MATERIAL SPECIFICATIONS.
QUALITY SYSTEM MEETS REQUIREMENTS OF DIRECTIVE 2014/68/EU (PRESSURE EQUIPMENT DIRECTIVE), ANNEX 1, CHAPTER 4.3 PER ABS GROUP LTD CERTIFICATE 41734 (EXPIRES JANUARY 28, 2027)

HUNTINGTON ALLOYS CORPORATION IS AN ACCREDITED INDEPENDENT NADCAP MATERIALS TESTING LABORATORY VIA CERTIFICATE NUMBER 3200230925 EXPIRATION DATE: NOVEMBER 30, 2025 FOR ALL TESTING SPECIFIED IN THE SCOPE OF ACCREDITATION.
MATERIAL PRODUCED UNDER QA SYSTEM DOCUMENTED IN HUNTINGTON ALLOYS CORP QA MANUAL REV. 56, DATED 01/25/2024
EN 10 204 / DIN 50049 (TYPE 3.1)
QUALITY SYSTEM CERTIFICATION: AS9100D:2016, EN9100:2018, JISQ9100:2016 and ISO9001:2015 (ABS-QE CERT 56476 EXPIRES 1/28/2027)
LABORATORY IS ACCREDITED TO ISO/IEC 17025:2017 FOR MECHANICAL TESTING AND CHEMICAL ANALYSIS.

INCOLOY®, INCONEL®, MONEL®, NILO®, NIMONIC®, Ni-SPAN-C®, UDIMET®, DURANICKEL®, 601GC®, 625LCF®, 718SPF®, 740H®, 800HT®, 945®, and 945X® are trademarks of the Special Metals group of companies EF595

HUNTINGTON ALLOYS CORPORATION
3200 Riverside Drive, Huntington, West Virginia 25705-1771 USA
Tel: +1.304.526.5100 Toll-Free in the USA: 1.800.334.4626
Fax: +1.304.526.5643 info@specialmetals.com

Certificate No. 146118-00
Dated 25-FEB-25
Page No. 8 / 8

CERTIFIED MATERIALS TEST REPORT

VISUAL AND DIMENSIONAL EXAMINATION SATISFACTORY.
MATERIAL, WHEN SHIPPED, IS FREE FROM CONTAMINATION BY MERCURY, RADIUM, ALPHA SOURCE, AND LOW MELTING ELEMENTS.

CHEMICAL ANALYSIS AS REQUIRED FOR CARBON, SULFUR, NITROGEN, OR OXYGEN IS PERFORMED BY COMBUSTION TECHNIQUES.
ALL OTHER REPORTED ELEMENTS ARE ANALYZED BY X-RAY AND/OR EMISSION SPECTROSCOPY.""

GE Transportation Aviation Special Process Certification Source Code 47150, Expiration Date of Dec 30, 2025

AUTHORIZED QUALITY CERTIFICATION REPRESENTATIVES:
D.L. CONLEY K.R. BREWER C.S. GUNNOE G.J. BURKHEAD T.A. WILKS

End Of Certificate

This is to certify that all required samplings inspections and tests have been performed in accordance with the order and specification requirements. The test report represents the actual attributes of the material furnished and the values shown are correct and true. The material described by this certificate is in full compliance with all order and inspection requirements. We hereby certify that the figures given are in accordance with the specified contract requirements.

REV. 8/08

Signed
For and on behalf of HUNTINGTON ALLOYS CORPORATION
Authorized Signature

INCOLOY®, INCONEL®, MONEL®, NILO®, NIMONIC®, Ni-SPAN-C®, UDIMET®, DURANICKEL®, 601GC®, 625LCF®, 718SPF®, 740H®, 800HT®, 945®, and 945X® are trademarks of the Special Metals group of companies

EF595

Customer PO#	Sales Order#	Howco Disp Cert#	Date	Authorized by	Page No.	Total Pages
TRP0049827	2272501-1	2272501-1	23-Apr-2025	Regina Euton	8	18

SPECIAL METALS

Ultrasonic Test Report

H-2245 Rev. 9/2023

Job Number	R3429191-01	**QC Procedure**	QCP 285 Rev 3 Date 03-03-2014
Heat Lot Number	LT7386EY11	**Additional Instruction**	**Supplement** **Rev** **Date** **Addendum** **Other** **Other**
Nom Diameter or X-section, wall or pipe schedule (in)	11.000"		
Number of Pieces **Piece Numbers if applied** **Serial Numbers if applied**	1 1	**Date of Inspection YYYY-MM-DD**	2025-02-18
Time of test, with respect to final length, width and heat-treat condition	Tested at final cross-section Tested prior to cutting final length / width Tested as final heat-treat condition	**Inspector and Certification Levels**	K. Gross SNT-TC-1ALevel II MIL-STD-2132 Inspector

Total Number of Probes Used for Test	1	**Shear Wave**				**Longitudinal Wave**		
		Circumferential Propagation		**Axial Propagation**		**Flaw Detection**	**Wall Thickness**	☐
		#1	**#2 Opposing**	**#1**	**#2 Opposing**	Contact Method	**Back Echo Monitor**	☐
Instrument Manufacture Model Serial Number Linearity Date m-d-y Frequency Filter Setting Cal Gain / Scan Gain dB		/	/	/	/	Staveley-Sonic 237 142PB 12-04-2023 2.25 MHz 50.0 / 56.0	/	
Transducer Manufacture Model Serial Number Frequency MHz Element Size (in) Focal Length (in) Effective Beam (in) Refracted Angle Cable length/Type (ft) Reject On / Off		/	/	/	/	Single Element Krautkramer GAMMA RHP 13E026XD 2.25 1.0 dia 0 7.5 / RG/174/U OFF	/	
Scanning Method Speed (ips) Index Type Max Index Used (in) Max Pulse Spacie (in) Couplant						Manual 6 Step Index Oil		
Reference Amplitudes % FSH % FSH Back Echo % Back Echo Loss EDAC Used						80 80 50 Yes		
Gating Delay and Range 1 Logic 1 % Ref Alarm Level 1 Delay and Range 2 Logic 2 % Ref Alarm Level 2						FS to BR Positive 50 BR Negative 50		
Reportable Conditions (Indications or BR loss)						None		
Remarks	1 WATER WASHABLE LP INSPECTION PERFORMED FOR UT SHEAR WAVE. 2. 3.							

Calibration Standard Data (all measurements in inches)

Calibration Standard		**NLOD Depth**	**NLID Depth**	**NTOD Depth**	**NTID Depth**	**FBH Diameter**	**Wall Thickness Reference 1**
SN	K367					0.250, 0.250, 0.250	
Standard Finish	No smoother than material	**NLOD Length**	**NLID Length**	**NTOD Length**	**NTID Length**	Metal Travel ☒ Drilled Depth ☐	**Wall Thickness Reference 2**
Material Finish	Less than 6.3 Micrometers					2.831, 5.567, 10.31	
Acoustic Comp Check	Long Wave Checked, 0 dB compensation required			Shear Wave , dB compensation required			

☐ **Minimum Wall Thickness Maintained** **Strip Chart Recorder Used** NA **Scanning Equipment**

☒ **Inspected Material Accepted** **Attachments**

"NOTE: The recording of false, fictitious or fraudulent statements or entries on this document may be punishable as a felony under Federal Statute."

SPECIAL METALS CORPORATION

3200 Riverside Drive

Huntington, WV «25707-1771 U.S.A.
304-526-5768 Cell 304-633-9143
www.specialmetals.com

TIMOTHY THACKER
NDE Manger

VISION RECORD

NONDESTRUCTIVE TESTING PERSONNEL

This is to certify that Kevin Gross has completed a vision screening which demonstrated the following results:

✓ **Natural** vision ability to read the Jaeger Number 1 line from the standard Jaeger test card at a distance of not less twelve inches.

___ **Corrected** vision ability to read Jaeger Number 1 line from the standard Jaeger test card at a distance of not less twelve inches.

Corrective aides used: a) glasses b) contacts c) both a and b d) either a or b

✓ Contrast discrimination **is sufficient** for requirements of NDT methods applied.

___ **Unable** to read Jaeger Number 1 line from the standard Jaeger test card at a distance of not less twelve inches. NDT certification at this time is not possible. Referred to eye care professional.

___ Contrast discrimination **is unacceptable** for requirements of NDT methods applied. NDT certification is not possible.

Date 01/27/2025

Timothy Thacker
NDE Manager,
Responsible Level III
ASNT NDT Level III UT, PT 216779
MIL-STD-2132 Test Examiner UT, PT, VT

NOTE: Ability to read the standard Jaeger "J1" test card at a distance of not less than twelve inches has been determined by the responsible Level III to be the equivalent of ability to achieve Snellen 20/25 rating.

INCONEL, INCOLOY, MONEL, NIMONIC, INCO-WELD, NILO, BRIGHTRAY, CORRONEL and UDIMET are registered trademarks of Special Metals Corporation

SPECIAL METALS
A PCC Company

Nondestructive Testing Personnel Certification

This is to certify that

Kevin Gross

has met all applicable training and examination requirements as outlined in MIL-STD-2132 and NSTR 99 and is therefore awarded certification as

Inspector, Ultrasonic – Immersion

Date of Certification: January 8, 2024

Date of Expiration: December 18, 2026
(Based on examination start date)

Vision Status: J1, Natural

T.P. Thacker,
Manager NDE
MIL-STD-2132 Test Examiner UT, PT, VT
ASNT NDT LEVEL III UT, PT 216779
SNT-TC-1A LEVEL III UT, PT

Corrective Aide:

Full documentation relating to this certification, which includes; training and experience histories, examination results and previous certification histories, is available for on-site review.

SPECIAL METALS
A PCC Company

Nondestructive Test Personnel Certification

This is to certify that

Kevin Gross

has met all applicable training and examination requirements as outlined in SMC written practice TQC 1 which meets or exceeds the requirements of SNT-TC-1A current and all prior editions, NAS 410 and T9074-AS-GIB-010/271.

Based upon training history, written, operational and physical examinations as required by the documents referenced above, this certification is awarded for

Level II, Ultrasonic-Immersion , Contact

Date of Issue: **January 8, 2024**

Date of Expiration: **December 18, 2026**

Vision Status: **J1, Natural**

T. Thacker Manager NDE
MIL-STD-2132 Test Examiner UT, PT, VT
ASNT NDT LEVEL III UT, PT 216776
SNT-TC-1A LEVEL III UT, PT

Correction Aid:

Documentation relating to this certification including examinations and previous certifications is available for on-site review.

Customer PO#	Sales Order#	Howco Disp Cert#	Date	Authorized by	Page No.	Total Pages
TRP0049827	2272501-1	2272501-1	23-Apr-2025	Regina Euton	12	18

SPECIAL METALS
A PCC Company

Nondestructive Test Personnel Certification

This is to certify that

Kevin Gross

has met all applicable training and examination requirements as outlined in MIL-STD-2132 and therefore, based on written, operational and physical examinations as specified in those documents referenced above, is awarded certification as

INSPECTOR, Visual, Dimensional and Alloy ID

Date of Issue: May 19, 2022

Date of Expiration: May 19, 2025

Vision Status: J1, Natural

T. Thacker Manager NDE
MIL-STD-2132 Test Examiner UT, PT, VT
ASNT NDT LEVEL III UT, PT 216776
SNT-TC-1A LEVEL III UT, PT

Documentation relating to this certification including examinations and previous certifications is available for on-site review.

Correction Aid: None

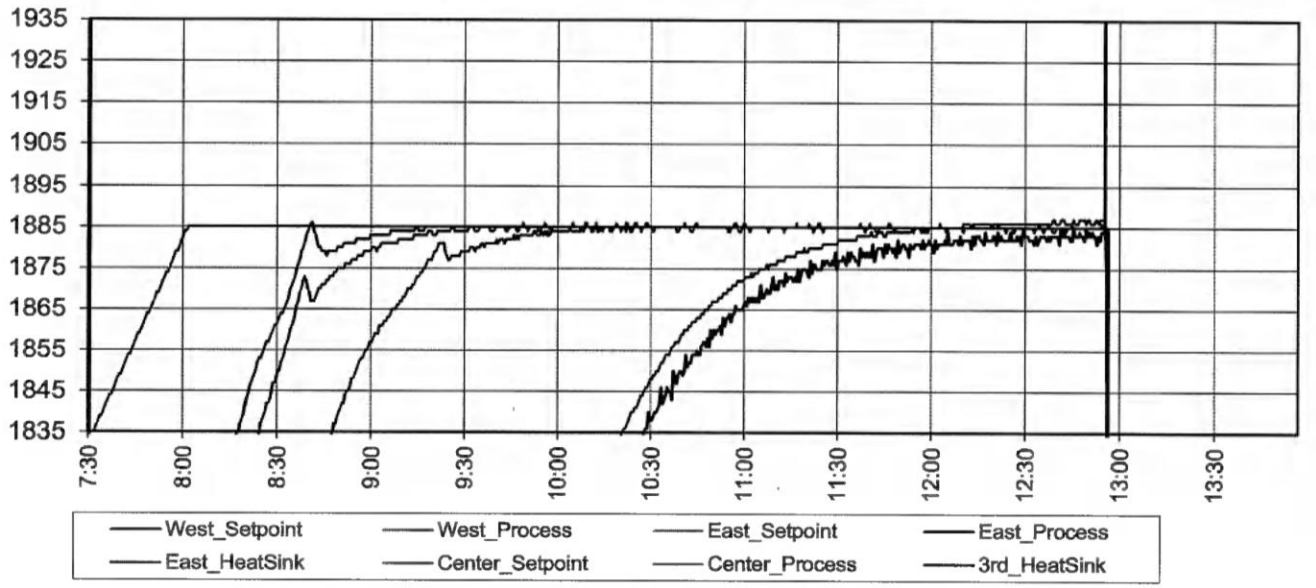
Customer PO#	Sales Order#	Howco Disp Cert#	Date	Authorized by	Page No.	Total Pages
TRP0049827	2272501-1	2272501-1	23-Apr-2025	Regina Euton	13	18

Order Report - Machine Shop Tip-Up Furnace #2 (50 Ft.)

Date & Time:	2/16/25	7:30:06
Operator:	13608	
Heat Type:	Solution Anneal	
Recipe #	167	
Quench:	Yes	
Water Temp (Start/End):		60 / 67
Maximum Water Temp:		67

Heat Sink Information	
East Heat Lot#	059142-24
East Heat Offset	1 °F
East heat Sink Size	Contact T/C
3rd Heat Lot#	086598-24
3rd Heat Offset	2 °F
3rd Heat Sink Size	Contact T/C

Event	Recipe Information			Event Information			
	Temp. SP	Hold Time (In Min.)	Ramp Rate (Per Hr.)	Start Date	Start Time	End Date	End Time
Charge				2/16/2025	7:30:06		
Step #1	1,885 °F	120	0 °F	2/16/2025	10:54:16	2/16/2025	12:54:16
Step #2	0 °F	0	0 °F				
Step #3	0 °F	0	0 °F				
Step #4	0 °F	0					
Discharge				2/16/2025	12:56:02		
Quench Time	60	Start Recipe HS Offset	25 °F	2/16/2025	12:56:24	2/16/2025	13:56:28



Order	Heat	Lot	Alloy	Weight	Dim1	Dim2	Dim3	Pieces	Location
R3429191-01	LT7386EY	11-1	718	7740	0	11.355	260.062	1	C11

THE RECORDING OF FALSE, FICTITIOUS OR FRAUDULENT STATEMENTS OR ENTRIES ON THIS DOCUMENT MAY BE PUNISHABLE AS A FELONY UNDER FEDERAL STATUTE.

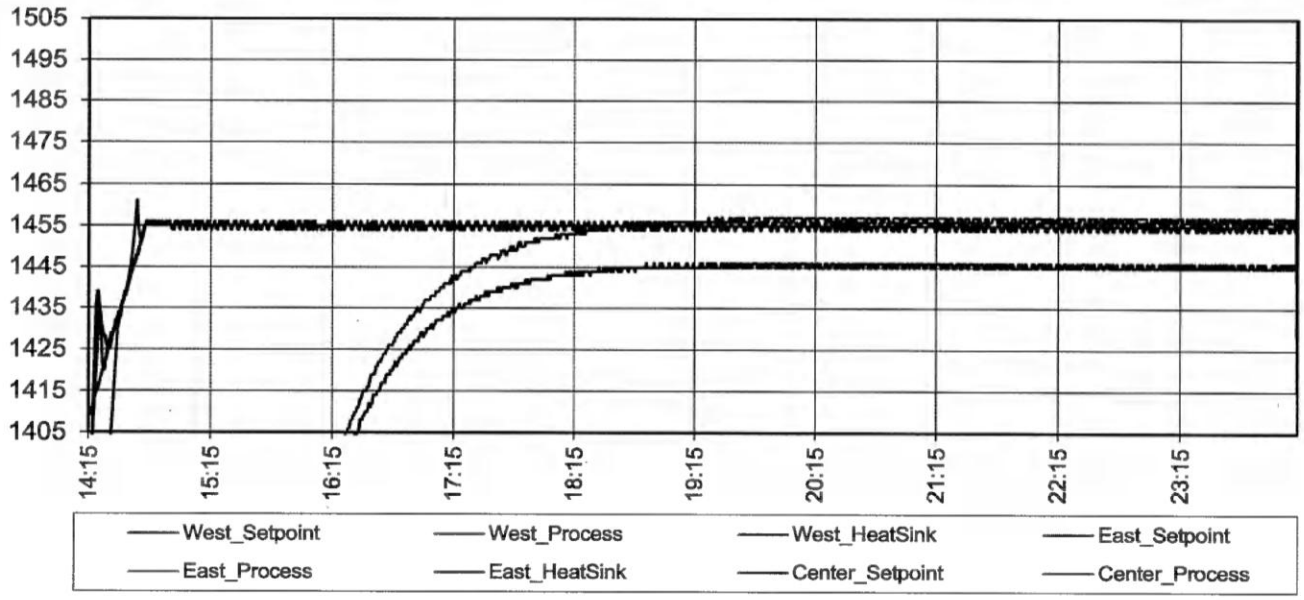
Customer PO#	Sales Order#	Howco Disp Cert#	Date	Authorized by	Page No.	Total Pages
TRP0049827	2272501-1	2272501-1	23-Apr-2025	Regina Euton	14	18

Order Report - Machine Shop Tip-Up Furnace #2 (50 Ft.)

Date & Time: 2/16/25 14:11:23
Operator: 13608
Heat Type: Age Hard
Recipe # 132
Quench: No
Water Temp (Start/End):

Heat Sink Information	
East Heat Lot#	059142-24
East Heat Offset	°F
East heat Sink Size	Contact T/C
West Heat Lot#	083462-24
West Heat Offset	2 °F
West Heat Sink Size	Contact T/C

Event	Recipe Information			Event Information			
	Temp. SP	Hold Time (In Min.)	Ramp Rate (Per Hr.)	Start Date	Start Time	End Date	End Time
Charge				2/16/2025	14:11:23		
Step #1	1,455 °F	390	0 °F	2/16/2025	17:40:26	2/17/2025	0:10:25
Step #2	0 °F	0	0 °F				
Step #3	0 °F	0	0 °F				
Step #4	0 °F	0					
Discharge				2/17/2025	0:12:10		
Quench Time		Start Recipe HS Offset	15 °F				



Order	Heat	Lot	Alloy	Weight	Dim1	Dim2	Dim3	Pieces	Location
R3429191-01	LT7386EY	11-1	718	7740	0	11.355	260.062	1	C11

THE RECORDING OF FALSE, FICTITIOUS OR FRAUDULENT STATEMENTS OR ENTRIES ON THIS DOCUMENT MAY BE PUNISHABLE AS A FELONY UNDER FEDERAL STATUTE.

Customer PO#	Sales Order#	Howco Disp Cert#	Date	Authorized by	Page No.	Total Pages
TRP0049827	2272501-1	2272501-1	23-Apr-2025	Regina Euton	15	18

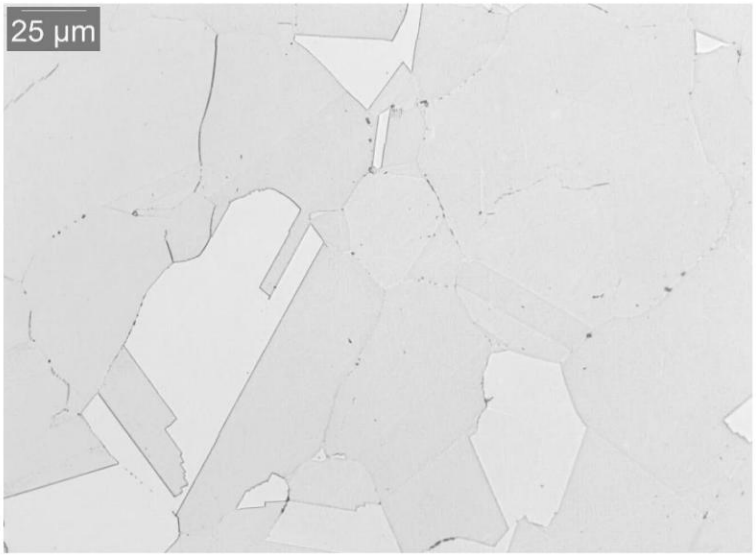


Figure 1: MS24273

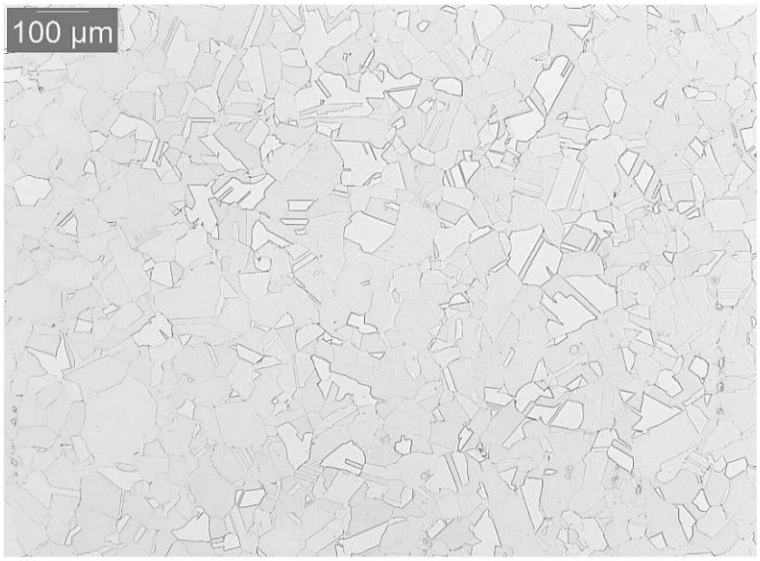


Figure 2: GS24274

SPECIAL METALS

Sales Order Number	300042919-1
Heat Number/Lot No.	LT7386EY11
Piece I.D.	1 TOE
Alloy	INCONEL alloy 718
Product Form	Round
Product Size	11.0000”
Product Condition	Solution Ann and Aged
Lab Heat Treat	None
Image Number	See Photomicrograph
Magnification	500x-100x
Orientation	Long-Mr
Etchant	7-Acids
Etch Time	Standard

NOTE: The recording of false, fictitious or fraudulent statements or entries on this document may be punishable as a felony under Federal Statute.

Customer PO#	Sales Order#	Howco Disp Cert#	Date	Authorized by	Page No.	Total Pages
TRP0049827	2272501-1	2272501-1	23-Apr-2025	Regina Euton	16	18

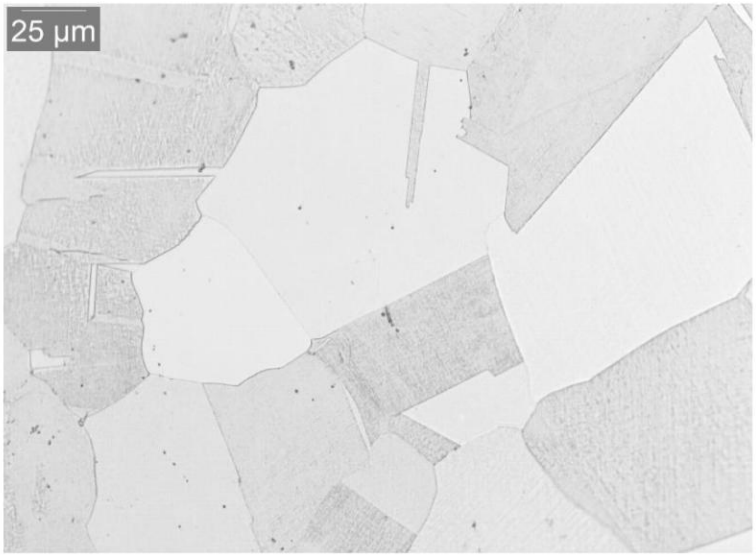


Figure 1: MS24271

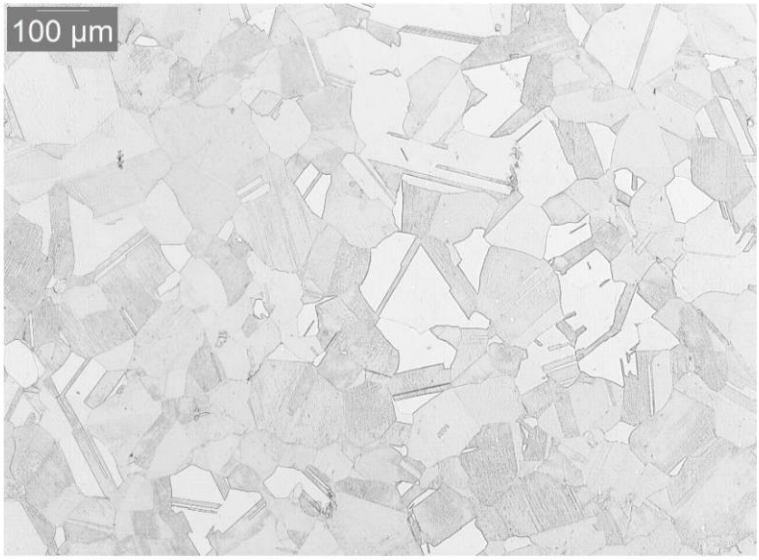


Figure 2: GS24272

SPECIAL METALS

Sales Order Number	300042919-1
Heat Number/Lot No.	LT7386EY11
Piece I.D.	1 TOE
Alloy	INCONEL alloy 718
Product Form	Round
Product Size	11.0000”
Product Condition	Solution Ann and Aged
Lab Heat Treat	None
Image Number	See Photomicrograph
Magnification	500x-100x
Orientation	Long-Sur
Etchant	7-Acids
Etch Time	Standard

NOTE: The recording of false, fictitious or fraudulent statements or entries on this document may be punishable as a felony under Federal Statute.

Customer PO#	Sales Order#	Howco Disp Cert#	Date	Authorized by	Page No.	Total Pages
TRP0049827	2272501-1	2272501-1	23-Apr-2025	Regina Euton	17	18

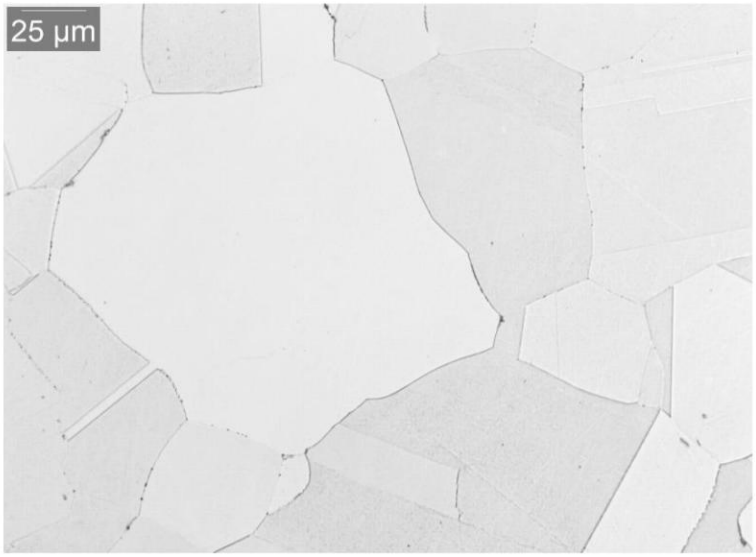


Figure 1: MS24275

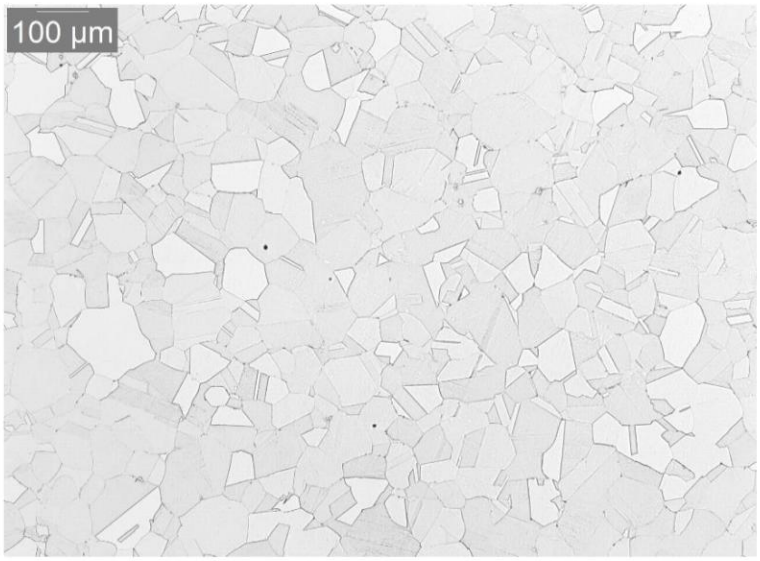


Figure 2: GS24276

SPECIAL METALS

Sales Order Number	300042919-1
Heat Number/Lot No.	LT7386EY11
Piece I.D.	1 TOE
Alloy	INCONEL alloy 718
Product Form	Round
Product Size	11.0000”
Product Condition	Solution Ann and Aged
Lab Heat Treat	None
Image Number	See Photomicrograph
Magnification	500x-100x
Orientation	Long-Cen
Etchant	7-Acids
Etch Time	Standard

NOTE: The recording of false, fictitious or fraudulent statements or entries on this document may be punishable as a felony under Federal Statute.

Customer PO#	Sales Order#	Howco Disp Cert#	Date	Authorized by	Page No.	Total Pages
TRP0049827	2272501-1	2272501-1	23-Apr-2025	Regina Euton	18	18

CACTUS WELLHEAD, LLC

SDR NO. 5939

SUPPLIER DEVIATION REQUEST (SDR)

Complete this form and email to the Cactus Wellhead Buyer stated on the Purchase Order. Remember to include a detailed deviation description and provide marked up drawings when applicable.

Supplier Name & Address:	*Purchase Order Number*	*Line Item*
Howco 9611 Telge Road Houston, TX	TRP0049827	Line 1 -13

Part Number	144712T	***Rev:***	A01	***Quantity:***	QTY 13 X 27.5"
Drawing Number:	n/a			***Rev:***	n/a
Part Description:	BAR, RND, 11 OD, INCONEL 718				
Heat Number(s):	LT7393EY11		***Serial Number(s):***		2272501

Deviation Description:

UT not performed per CNT-005, request approval to accept mill UT reports

Submitted By:	James Andrews	***Date:***	05/29/25

Cactus Wellhead Disposition Only

Accept (Justify Below)	No. of Pcs. 13 X 27.5"	**Rework**	No. of Pcs.	**Scrap**	No. of Pcs.

Engineering		***Quality***	
Signature	Date 4/29/2025	Signature	Date 4/29/2025

ACCEPT JUSTIFICATION:

QA approved mill UT.

REWORK ROUTING:

Supplier Action Required: ***Material shall be tagged or marked with the SDR number and a copy of the SDR shall be included with the material certifications and/or bill of lading. If material is not delivered in accordance with these instructions, material will be subject to rejection and processed as a Supplier Nonconformance.***